Introduce yourself in the chat!

Tell us your program and concentration

Hello there

I'm Omer, your instructor for today!

- Studied Management Science and Computer Science in my undergrad
- Graduated from the MI program at the iSchool where I studied Data Science
- Currently working for the Ontario Ministry of Children Community and Social Services as a Data Scientist
- And I love teaching!

Housekeeping

- This session is interactive! Please ask questions when prompted, I’ll be stopping regularly to check if everyone is understanding the concepts
- We will have a short break in the middle (stretch, grab some tea)
- In case, we aren’t able to cover something properly, we will skip it. The goal is to learn the content, not rush through
- We will be doing some coding together
    - Sometimes you’ll code and we then go through the solution together!
- You will have access to the slides throughout in case you need to go back to a particular slide

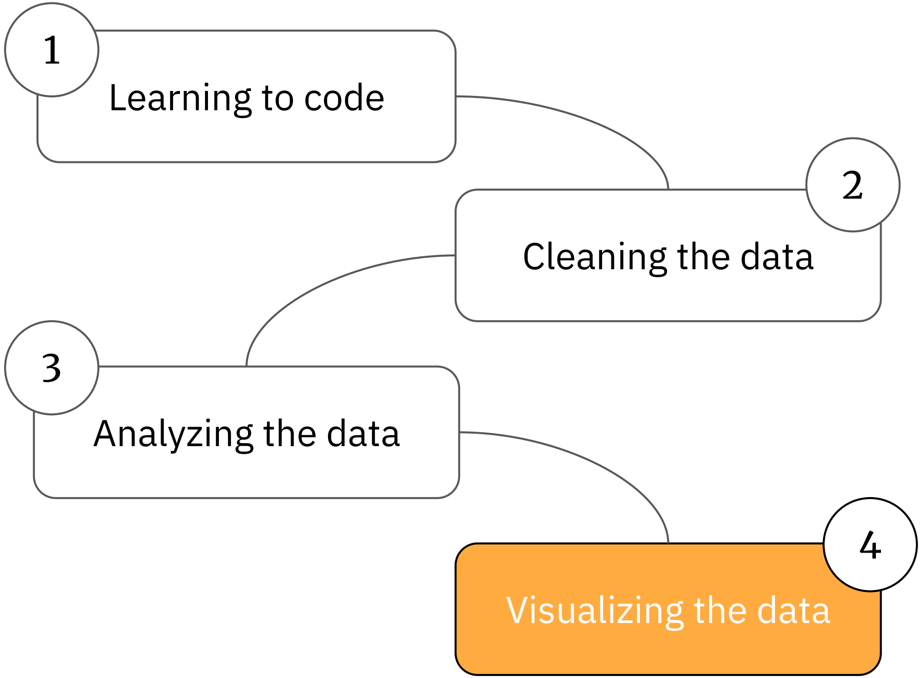
Here’s What We’ll Cover

1. Bad data visualizations
2. What makes a good data visualization
3. Plotting with matplotlib
4. How to keep learning

Prologue

What we’ve studied

Our journey


1
Learning to code
2
Cleaning the data
3
Analyzing the data
4
Visualizing the data

Visualizations surround us

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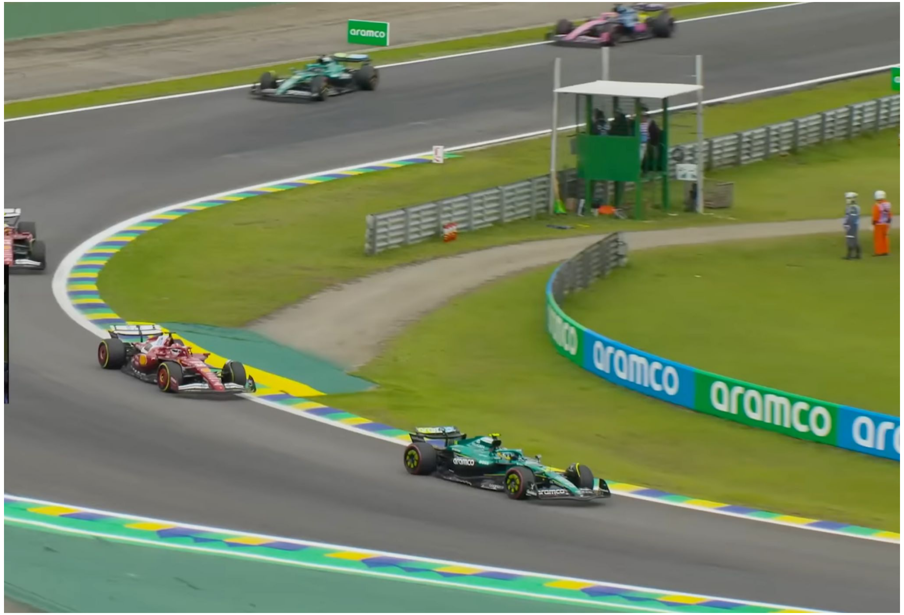

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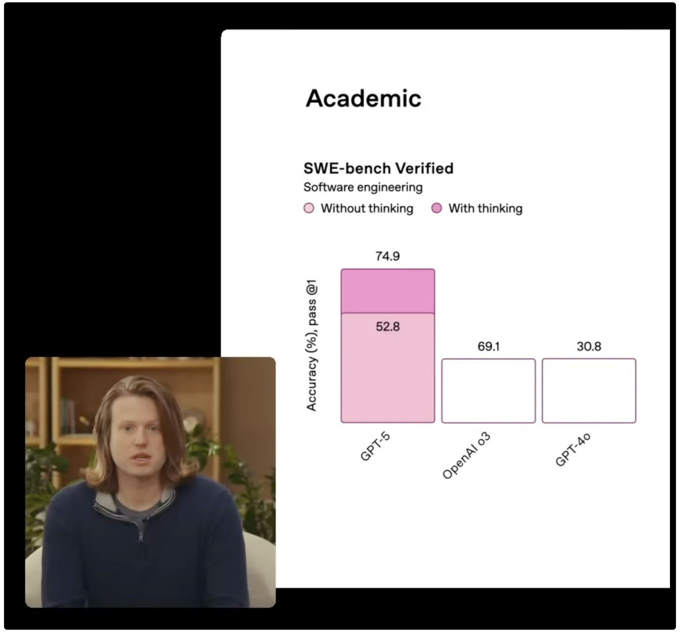
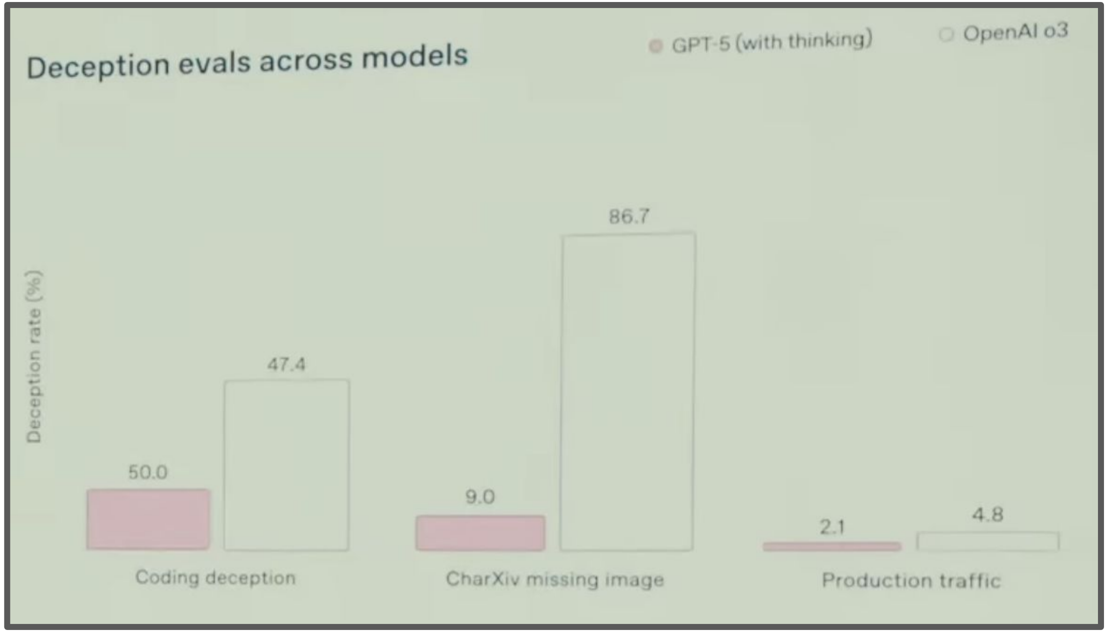
Chapter 1

Bad data visualizations

Malicious





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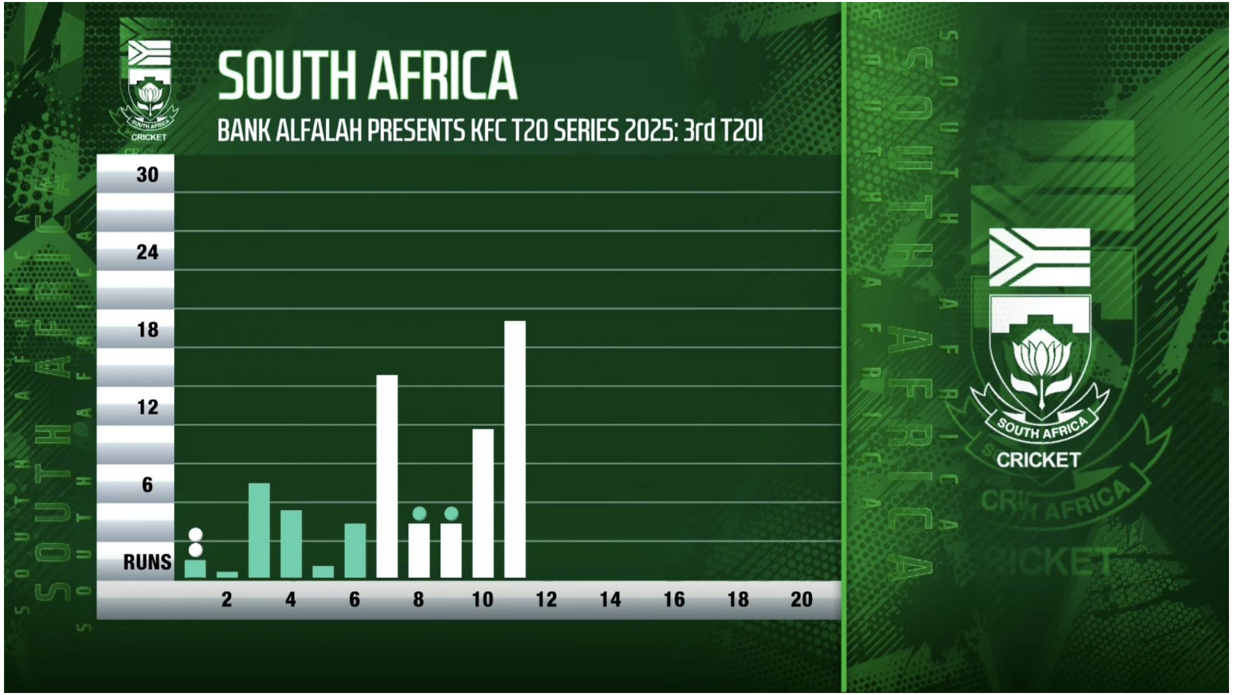
Busy



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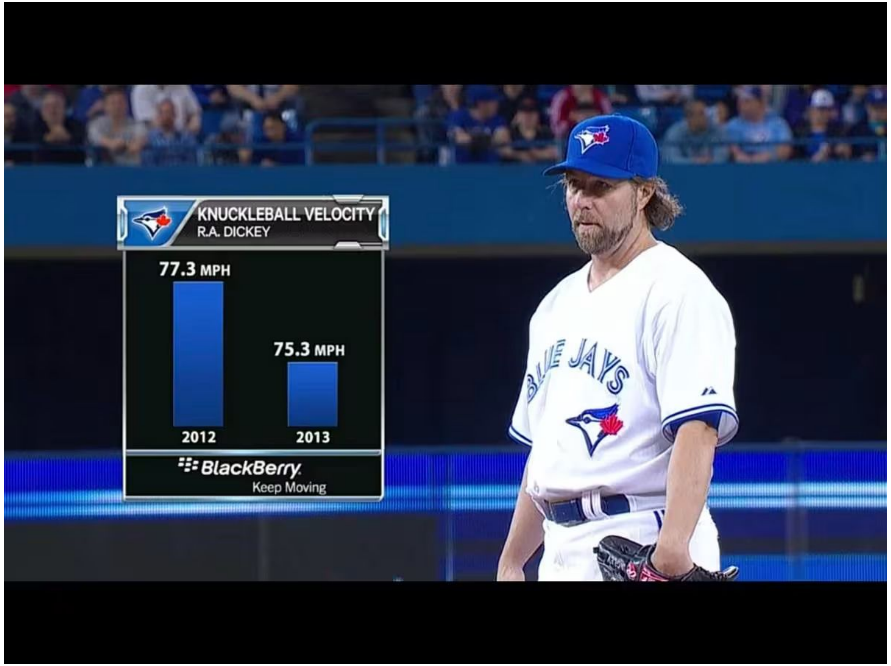
Misleading


S&P 500
Total Job Openings
ChatGPT
Launch
2005
2007
2009
2011
2013
2015
2017
2019
2021
2023
2025

Unfair



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The WHITE HOUSE
U.S. STEEL PRODUCTION INCREASES
Total U.S. Steel Production (Mt) 2024 v. 2025
81.8
81.6
81.4
81.2
81.0
80.8
80.6
80.4
80.2
2024 Total Steel Production (Mt)
Total Steel Production 2025 (Mt)

Source

Chapter 2

What makes a good data visualization

Some data viz greats

Source

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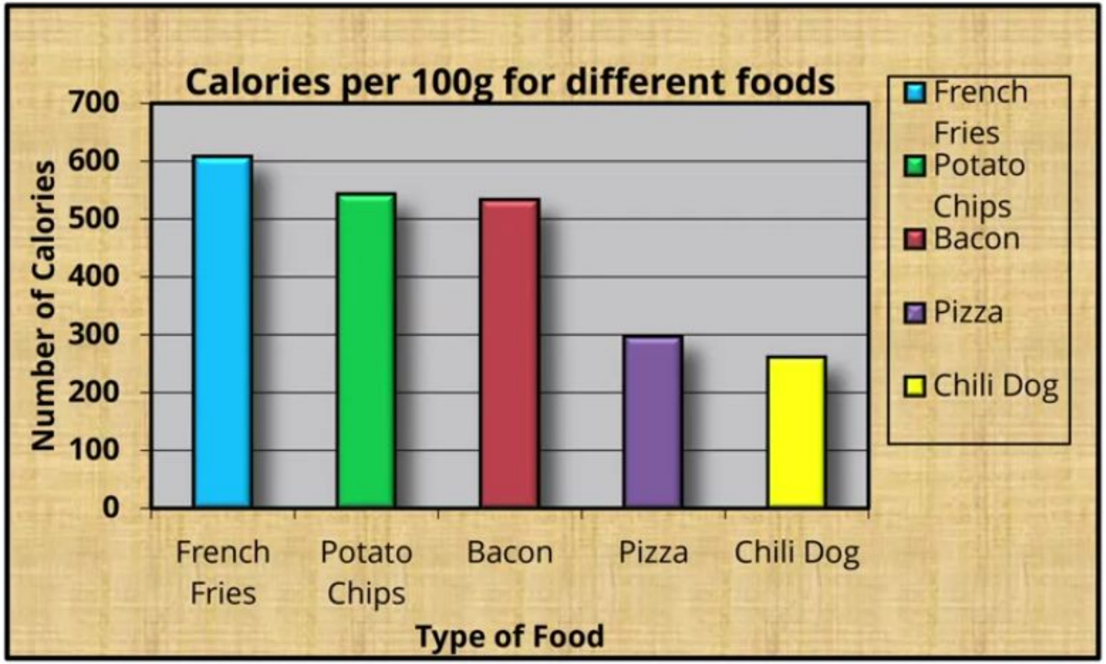
Visualization principles

Here are some general suggestions to keep in mind when visualizing data,

1. Minimize chartjunk – remove elements that don’t convey insights (GeeksForGeeks)
2. Ensure accessibility – keep your visualizations inclusive for all people (GeeksForGeeks)
3. One point per chart – use phrasing such as “the data indicates…” (Schwartzberg)
4. Highlight important points – mark portions of the graph that are critical (Schwartzberg)
5. Label well – don’t leave anything for the audience to guess (Schwartzberg)
6. Use explanatory titles – write the insight as the title instead of a description alone (Dkyes)

Too much Chartjunk



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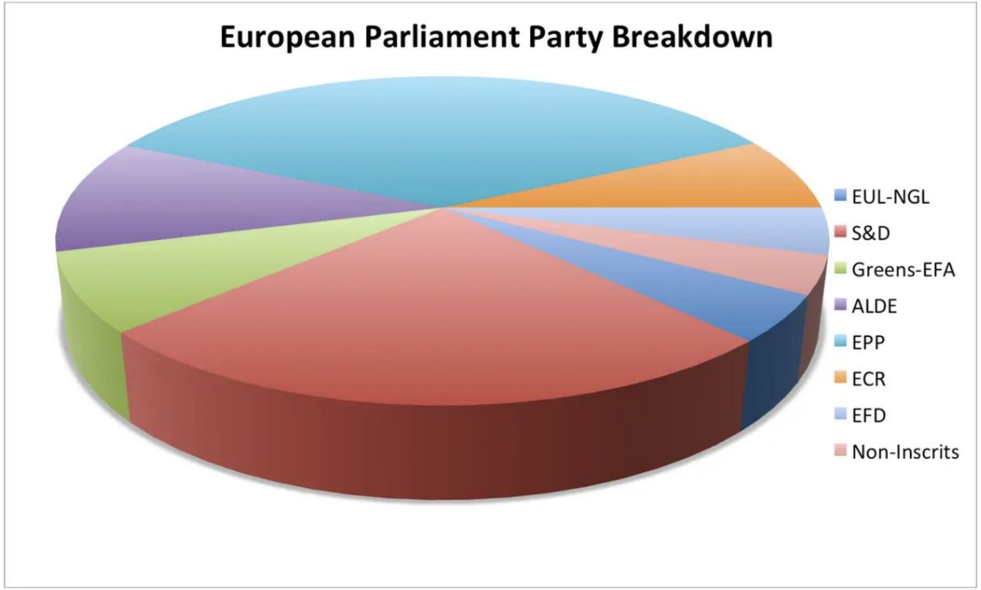
Let’s look at a transformation

Pie Chart = Bad



Walter Hickey / BI

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Let’s take 10

Chapter 3

Plotting with matplotlib and seaborn

The essentials

There are a lot of graphs to choose from. But we can get most of our work done with simply using these two,

- Bar plots are used for comparing values of across categories
- Line plots are used for showing change over a period of time

We'll be using Matplotlib and Seaborn

- Matplotlib is a library for visualizing data
  - more low-level (you can customize much more)
- For our needs, we'll be using Seaborn
  - also a library for visualizing data with much easier parameters (high-level)

Exercise time!

We are going to be using Jupyter Notebook for our coding needs today. It’s an open-source web-application for writing computational documents. Let’s get started!

- Open jupyter.org/try
- Click on JupyterLab
- Let’s create a new notebook
    - Click on File -> New -> Notebook
    - Click next with the default settings
    - We will need to install seaborn
        - For this, run %pip install seaborn in a code cell

Getting started

- Let's first load those libraries up
- We'll practice with the same datasets used for Python 103

```
import matplotlib.pyplot as plt
import seaborn as sns
```

A bar plot

You need to specify the following:

1. data - this should be your dataframe
2. x - what value goes to the horizontal axis
3. y - what value goes to the vertical axis
4. estimator - the statistic you want to use (defaults to mean)
5. palette - your choice of colors

```
sns.barplot(data=df, x="sex", y="weight", estimator = "median", palette = "pastel" )
```

A line plot

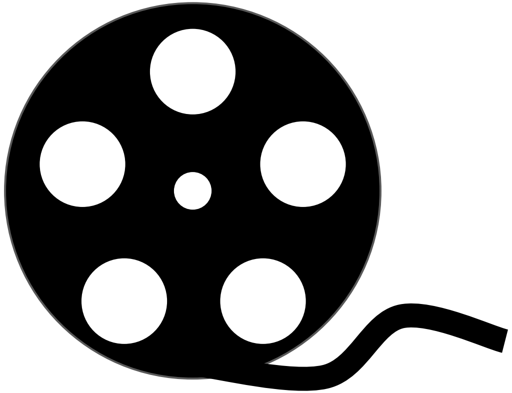
All the same variables you find in the barplot plus:

- markers - what object to mark the events in the plot

sns.lineplot(data=df, x=”year”, y=”weight”, estimator = “mean”, palette = “pastel”, markers = “o”)

Let’s test what we’ve learned

We’ll design a barplot and a lineplot with Gross (Adjusted), Year, and Studio variables

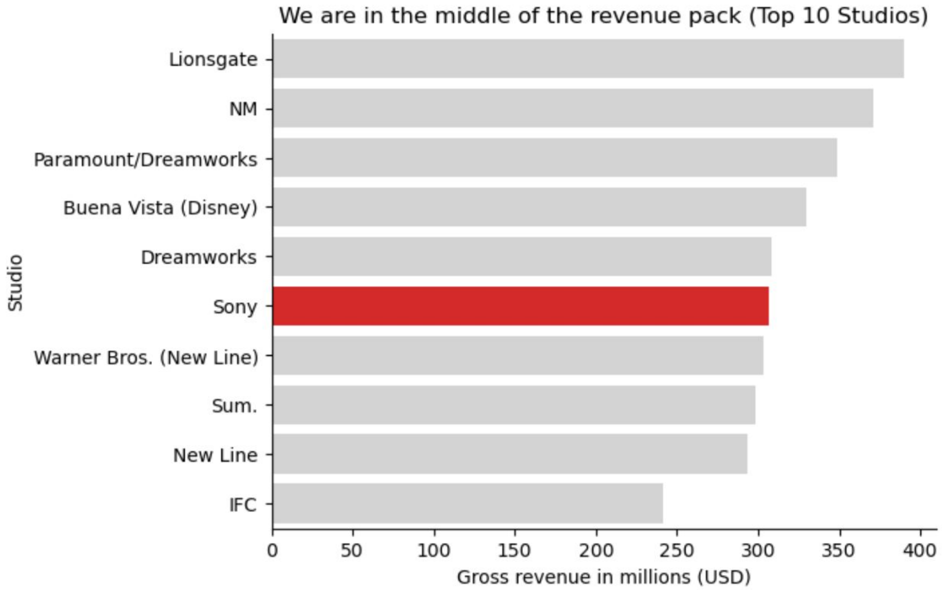


Things we'll be tweaking

- If you're doing EDA, then you don't need to concern yourself with the following
- Otherwise, if you're making the visual as part of your narrative, then we should make it according to the principles we discussed
  - For this we use the following attributes for a graph in Matplotlib:
    - plt.title() – the title of the graph
    - plt.ylabel() – the label for the vertical axis
    - plt.xlabel() – the label for the horizontal axis
    - plt.grid(axis="", linestyle="", alpha=) – for determining direction, form, and transparency of the grid lines
    - sns.despine() – for removing borders

Our revenue is competitive

- Over the last 40 years, Sony has remained in the top 10 studios
- We’ve earned on average 312 million USD globally



Exercise 3.1

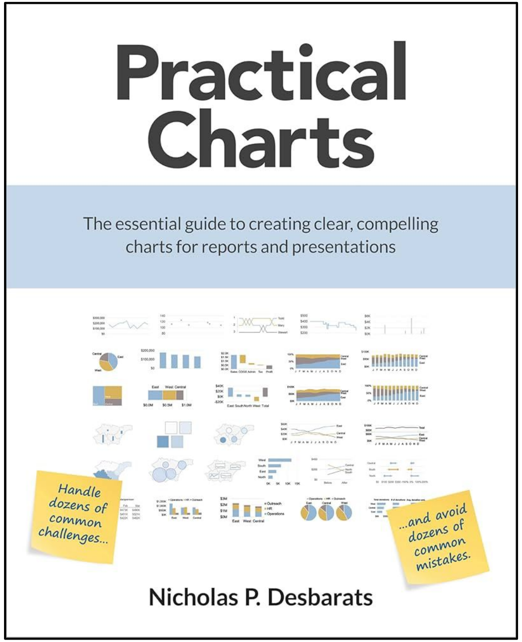
Time for you to practice. Use the surveys.csv file and make me visual. Whatever you want to make. Present me the story

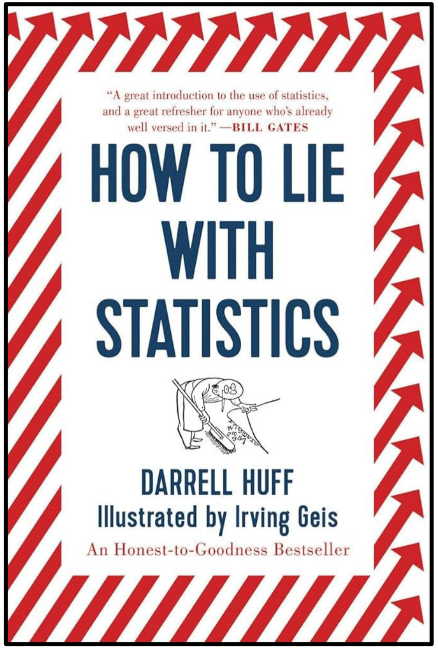
Bonus

How to keep learning

More Practice

- Follow data visualization experts online such as
  - Brent Dykes
  - Morgan Depenbusch
  - Nicola Rennie
- Read and apply learnings
- Take datasets and make visualizations for your portfolio
- Seek feedback on your work
- Recommended readings →





Thank you for tuning in!

I am always available for a coffee chat. Drop me a message on LinkedIn (Scan the QR code).

I would love to hear some feedback on the workshop. Please fill out this feedback survey from the iSchool.

Thank you for tuning to all the workshops! <3

References

Hickey, Walt. "The Worst Chart in the World." *Business Insider*, 17 June 2013, www.businessinsider.com/pie-charts-are-the-worst-2013-6

Schwartzberg, Joel. "Present Your Data Like a Pro." *Harvard Business Review*, 14 Feb. 2020, hbr.org/2020/02/present-your-data-like-a-pro

"Mastering Tufte's Data Visualization Principles." *GeeksforGeeks*, 21 June 2025, www.geeksforgeeks.org/data-visualization/mastering-tuftes-data-visualization-principles

Dykes, Brent. *Many data professionals underestimate the narrative power of explanatory headlines. Too often, the descriptive chart titles used in dashboards or...* 23 Oct. 2025, www.linkedin.com/feed/update/urn:li:activity:7387167142514917376

References

This is an excellent article by Nathan Yau on how to spot visualization lies

The surveys.csv file is from Portal Project Teaching Database

https://viz.wtf is an interesting list of visualizations that make no sense

I try to design my content in line with my Instructor training from the Carpentries